Torsion-coordinate DLP over quadratic extensions

Eight-arm same-target comparison with a public scaling-isomorphism control · 2026-10-14

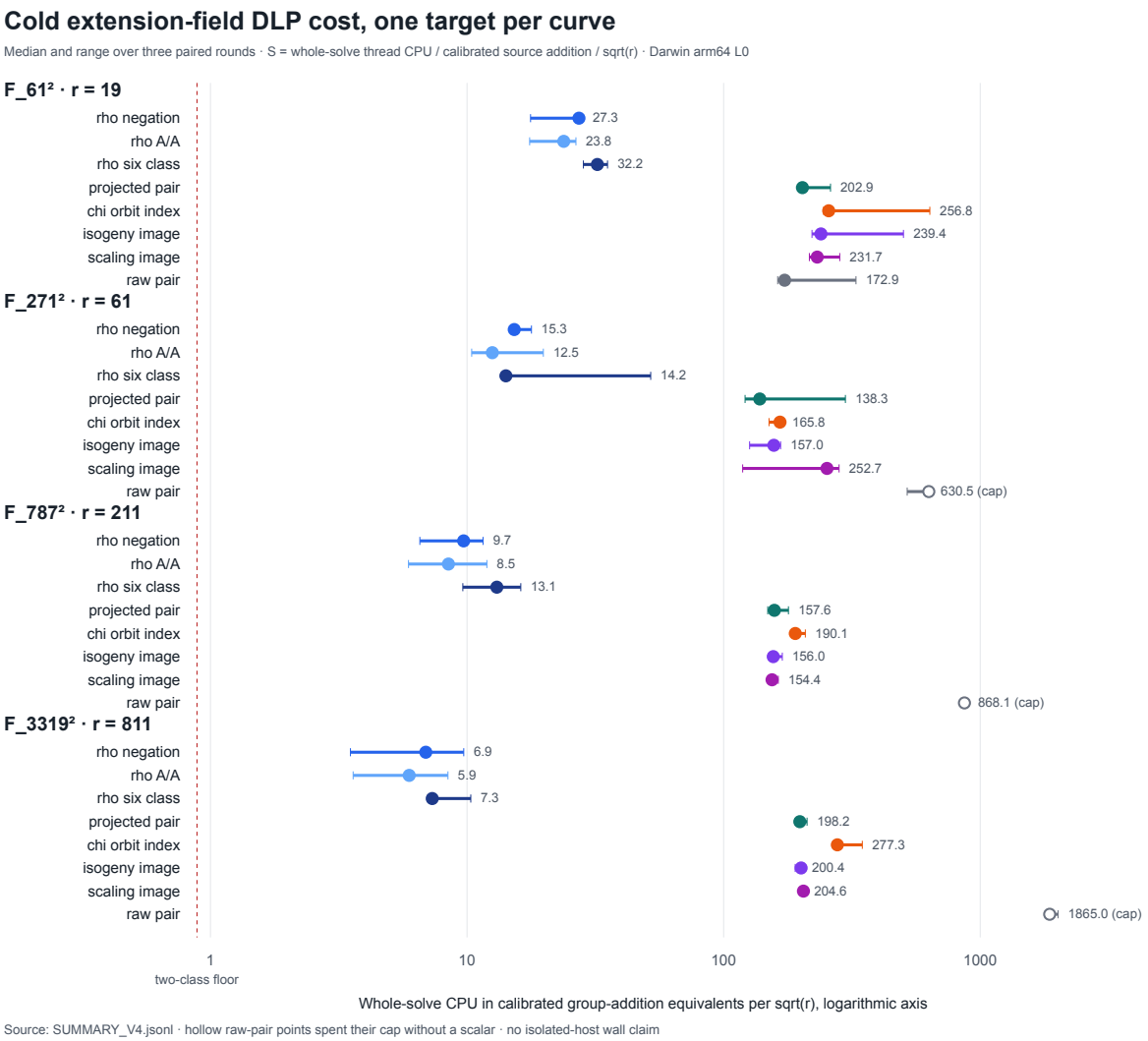
The frozen groups are prime-order subgroups of $E(F_{p^2})$, with $E : y^2 = x^3 + 1$ and orders 19, 61, 211, 811. Translation by rational three-torsion acts on $\chi = \frac{y+1}{y-1}$. The direct degree-three image is $E' : y^2 = x^3 - 27$. In every exact field tested, $u^2 = -3$ has a base-field solution, so $\psi_{u(x,y)} = (u^2 x, u^3 y)$ is also a public map to E'. The selected values of u were 27, 57, 28 and 264, each checked on 64 point pairs plus all frozen inputs.

The table uses the **whole cold current-thread CPU interval**, divided by measured source-addition time and $\sqrt{r}$, denoted $S_{\mathrm{CPU/GAE}}$. It charges setup, queries, relation recovery, map work and scalar verification. Every value is a three-round median on a shared Darwin arm64 host (L0); the χ/rho and scaling/rho columns are medians of paired full-cost ratios.

Order r	Projected	χ index	Degree-three	Scaling	Rho	χ/rho
19	202.85	256.79	239.44	231.65	27.30	14.53
61	138.26	165.85	156.96	252.68	15.28	11.09
211	157.59	190.15	156.02	154.41	9.70	21.46
811	198.17	277.34	200.41	204.56	6.91	50.27

The four projected arms had **identical relation count, query count, rank and recovered scalar in all twelve paired size-by-round cells**. The 96 run rows have 87 scalar recoveries checked by a second native F_{p^2} and curve-arithmetic implementation; nine raw-pair rows reached their cap. The χ **orbit index** increased median CPU over the plain projected table at every size. This result says nothing about the still-unimplemented χ polynomial decomposition oracle. The three-round ranges, including a wide scaling-arm spread at order 61, prevent map-specific wall-performance attribution.

Full arm ranges



Dots are medians; whiskers span the three measured rounds. Hollow raw-pair points are capped work with no scalar answer. The dashed generic counted-operation floor is a reference marker; the CPU-equivalent axis does not make it a universal lower bound in CPU time.

Accounting, proof and open work

The cold interval includes line lifting, square roots, cofactor projection, factor-base construction, pair-table allocation, failed and successful queries, exact source relation checks, modular elimination, target recovery and internal verification. The degree-three and scaling arms charge their map work; scaling constructs u by an ascending search in measured setup. Six-class rho charges eigenvalue discovery. Calibration and output serialization are outside method intervals. A fixed two-class generic counted-operation floor is $\sqrt{\frac{\pi}{4}} = 0.886226925$ per $\sqrt{r}$; converting CPU to local addition equivalents does not create a universal lower bound. Per-arm peak RSS, portable primitive conversion, L2 isolation and independent-host timing remain open.

The exact raw torsion-shift census found two extra subgroup support points at $p = 787$, but no extra cofactor-projected pair sum and no representation of that frozen target. The projected support sizes were 18/18, 58/60, 196/210 and 742/810 nonidentity subgroup points. These are finite observations on four frozen bases. Four exact coefficient witnesses are proven in a pinned Lean 4 build with no axioms. The universal sixth-power factorization is also proven in Lean using its standard foundations and checked by a separate native identity certificate. The prior conditional quotient-support theorem and polynomial χ certificates are linked from the report. A faithful curve-level Lean isomorphism and χ theorem is still open.

For this fixed source and destination, the scaling isomorphism exists over every F_{p^2} with odd $p \neq 3$: each nonzero $a \in F_p$ has order dividing $p - 1$, which divides $\frac{p^2-1}{2}$, so a is a square in F_{p^2}. Taking $a = -3$ gives the required u. The next non-isomorphic search must therefore use a different curve/isogeny pair. The finite-field existence theorem is written in the Markdown report but is not yet formalized in Lean.

The embedding degrees $\mathrm{ord}_{r(p)}$ were 9, 10, 105 and 810, or 9, 5, 105 and 405 relative to p^2. The root-of-unity output fields have those first four degrees over F_p, while common fields containing the frozen F_{p^2} points and the roots have degrees 18, 10, 210 and 810. A second pairing input may require more. A pairing transfer and multiplicative extension-field DLP were screened by those degrees only: they were neither constructed nor priced. The table instead compares complete elliptic-curve DLP methods on the same extension-field groups.

The corrected seven-arm V3 run and its 75-scalar replay remain in `REPORT_V3.md` and `REPORT_V3.pdf`. The final eight-arm bytes are in `RUN_V4.jsonl`; `REPLAY_RUN_V4.json` contains the independent 87-scalar check, `PAIRED_AUDIT_V4.jsonl` the twelve exact outcome comparisons, `SUMMARY_V4.jsonl` all 32 table rows, and `ARTIFACTS.sha256` the full manifest. The pre-merge release-library suite passed 4,009 tests with zero failures and 113 ignored; the merged PR tree passed 4,083 with zero failures and 117 ignored. Earlier failed environment receipts are retained. The Markdown `REPORT.md` gives the complete cost table, failures, proof scope and next gate.